AI-ENABLED HYBRID LIPO–LI-ION POWER MANAGEMENT MONITORING SYSTEM FOR FPV QUADCOPTER - SIMULATION

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Internship / Term Project Report
Trimester 7

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ABSTRACT

This report presents the design and simulation of an AI-enabled hybrid power management system for FPV quadcopters utilizing Lithium Polymer (LiPo) and Lithium-ion (Li-ion) batteries. The primary challenge addressed is the efficient, safe, and adaptive utilization of heterogeneous energy sources under highly dynamic load and mission conditions, where conventional single-battery systems are inherently limited in performance and reliability. This problem is critical for enabling extended flight endurance and robust autonomous operation in modern UAV applications.

The proposed approach integrates nonlinear battery modeling, real-time flight and environmental simulation, and an AI-enabled rule-based expert system for intelligent battery selection. The system incorporates key AI features, including context-aware decision-making based on multi-parameter inputs (load current, state of charge, voltage, and temperature), mission-aware control across different flight phases, predictive return-to-base triggering, reserve energy management for safe landing, hysteresis-based switching stability, and autonomous emergency landing decisions. These features enable adaptive and goal-oriented energy management in real time. The system is implemented in MATLAB with realistic flight profiles and environmental disturbances.

Experimental results demonstrate improved energy efficiency through optimal utilization of Li-ion during low-load conditions and LiPo during high-power and landing phases. The system also enhances operational safety and reliability, validating its effectiveness as an intelligent UAV power management solution.

Index Terms— FPV Quadcopter, Hybrid Battery Management, AI-Enabled Systems, Rule-Based Expert System, Autonomous Decision-Making, UAV Simulation

1. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), particularly First Person View (FPV) quadcopters, have gained significant importance in applications such as aerial surveillance, inspection, mapping, and high-speed racing. These systems operate under stringent energy constraints, where efficient power management is essential to ensure sustained performance, flight stability, and operational safety. The performance of an FPV drone is highly dependent on its battery system, as energy availability directly influences flight duration, maneuverability, and mission success.

Conventional UAV platforms typically rely on a single battery type, most commonly Lithium Polymer (LiPo) batteries, due to their high discharge capability. However, LiPo batteries exhibit limitations in terms of energy density and thermal sensitivity. In contrast, Lithium-ion (Li-ion) batteries offer higher energy density and longer endurance but are not suitable for high current demands encountered during aggressive maneuvers. This inherent trade-off between power capability and energy efficiency motivates the need for a hybrid battery management approach.

Efficiently integrating multiple battery types introduces several challenges, including real-time decision-making under dynamic load conditions, maintaining safety under voltage and temperature constraints, and adapting to different mission phases. These challenges are further amplified in autonomous UAV systems, where manual intervention is not feasible. Therefore, intelligent energy management mechanisms are required to enable adaptive, reliable, and safe operation.

In this work, an AI-enabled hybrid power management system is developed for FPV quadcopter simulation. The system combines LiPo and Li-ion batteries and employs an intelligent rule-based decision framework to dynamically allocate power based on real-time system states such as load current, state of charge, temperature, and mission context. A MATLAB-based simulation environment is used to model realistic flight profiles, environmental disturbances, and battery behavior, along with a real-time telemetry dashboard for monitoring system performance.

The remainder of this report is organized as follows. Section 2 describes the problem formulation and objectives. Section 3 details the methodology and system design. Experimental setup and results are presented in Section 4, followed by discussion in Section 5. Section 6 concludes the report and outlines future work.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

Efficient energy management is a critical challenge in FPV quadcopter systems due to the highly dynamic nature of flight operations. UAVs experience significant variations in load demand across different mission phases such as takeoff, cruising, acrobatic maneuvers, return-to-base, and landing. Conventional systems typically rely on a single battery type, which results in suboptimal performance because no single battery can simultaneously provide high power output, high energy density, and thermal stability.

Lithium Polymer (LiPo) batteries are well-suited for high current discharge but are limited by lower energy density and increased

thermal sensitivity. In contrast, Lithium-ion (Li-ion) batteries offer higher energy density and longer endurance but are not capable of sustaining high current demands required during aggressive flight conditions. This inherent trade-off leads to inefficient energy utilization and potential safety risks when a single battery type is used.

The problem addressed in this work is the design of an intelligent hybrid power management system that can dynamically allocate power between LiPo and Li-ion batteries in real time. The system must operate under multiple constraints, including nonlinear battery characteristics, voltage and temperature safety limits, and rapidly changing load conditions. Additionally, it must account for mission-specific requirements and ensure safe operation during critical phases such as return-to-base and landing.

The scope of this work is limited to simulation-based evaluation using a MATLAB environment, where realistic flight profiles, environmental disturbances, and battery behaviors are modeled. The system assumes accurate measurement of state variables such as current, voltage, temperature, and state of charge, and focuses on decision-making at the system level rather than hardware implementation.

2.2. Objectives

The primary objectives of this project are as follows:

1. To analyze the limitations of single-battery UAV systems and the advantages of hybrid energy configurations
2. To develop mathematical models for LiPo and Li-ion batteries, including voltage, state of charge, and temperature behavior
3. To design and implement an AI-enabled rule-based decision framework for dynamic battery selection
4. To simulate realistic drone flight conditions, including varying load profiles and environmental disturbances
5. To incorporate safety mechanisms based on voltage thresholds, temperature limits, and state of charge constraints
6. To implement mission-aware energy management, including return-to-base and emergency landing strategies
7. To develop a real-time telemetry dashboard for monitoring battery states, system performance, and decision outputs
8. To evaluate the effectiveness of the proposed system in terms of energy efficiency, operational safety, and system robustness

3. METHODOLOGY / APPROACH

This section describes the design and implementation of the AI-enabled hybrid power management system for FPV quadcopter simulation. The methodology integrates flight dynamics simulation, battery modeling, and an intelligent decision framework to enable real-time adaptive energy management.

3.1. Overall Workflow

The overall workflow of the proposed AI-enabled hybrid power management system is illustrated in **Fig. 1**. The system follows a structured pipeline that begins with input generation and progresses through simulation, modeling, intelligent decision-making, and visualization.

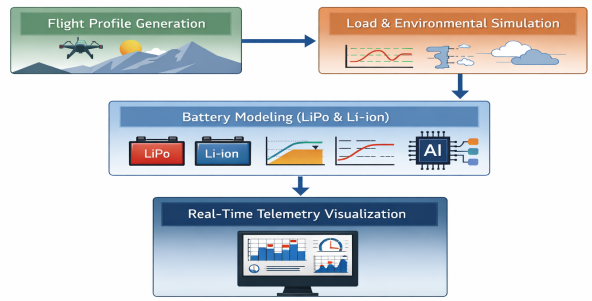

Fig. 1: Overall System Workflow

Fig. 1. Overall System Workflow

The process starts with flight profile generation, where realistic drone motion and mission phases such as takeoff, climb, cruise, and maneuvering are defined over time. These profiles determine the operational conditions under which the system is evaluated.

Next, the load and environmental simulation module computes the corresponding current demand and incorporates external disturbances such as wind. These factors introduce variability in system behavior and directly influence power consumption.

The battery modeling stage evaluates the state of both LiPo and Li-ion batteries using nonlinear mathematical models. This includes estimation of voltage, state of charge (SOC), and temperature, providing an accurate representation of battery dynamics under varying load conditions.

The computed battery states are then processed by the AI-enabled decision engine, which performs real-time battery selection based on multiple system parameters, including load current, SOC, voltage, temperature, and mission phase. This enables adaptive and context-aware power allocation.

Finally, the real-time telemetry visualization module presents system outputs through an interactive dashboard, displaying battery states, active power source, current draw, temperature, and flight information for monitoring and analysis.

As shown in **Fig. 1**, this end-to-end workflow ensures seamless integration of simulation, modeling, and intelligent decision-making for efficient and safe UAV power management.

3.2. Technical Details

This section presents the detailed implementation of the proposed system, including flight simulation, battery modeling, and the AI-enabled decision framework. The design emphasizes computational efficiency, realism, and modularity to ensure reproducibility.

A. Flight and Load Simulation: The system employs a discrete-time simulation with a fixed time step to model drone operation over a predefined mission duration. The flight profile is divided into multiple phases, each characterized by distinct load requirements:

- **Takeoff and climb:** High current demand due to thrust requirements
- **Cruise:** Moderate to low current with periodic variations
- **Acrobatic maneuvers:** Rapid fluctuations and peak current loads
- **Extended flight:** Stabilized but dynamic load behavior

The load current is modeled as a combination of deterministic patterns and stochastic variations to simulate real-world uncertainties. Environmental effects, particularly wind disturbances, are incorporated as random perturbations that influence both drone motion and energy consumption.

This approach ensures that the system is evaluated under realistic and dynamically varying conditions.

B. Battery Modeling: The system models two distinct battery types with complementary characteristics:

- **LiPo battery:** High discharge capability, lower energy density
- **Li-ion battery:** High energy density, lower discharge capability

To capture realistic behavior, nonlinear models are used for voltage, state of charge, and temperature.

State of Charge (SOC)

$$SOC = \frac{C_{remaining}}{C_{total}} \times 100 \tag{1}$$

LiPo Voltage Model

$$V_{LiPo} = 10.5 + 2.1SOC + 0.2\sqrt{SOC} - 0.5(1 - SOC)^3 \tag{2}$$

Li-ion Voltage Model

$$V_{Li-ion} = (3.0 + 1.2SOC + 0.15\log(1 + 9SOC)) \times N_{cells} \tag{3}$$

Discharge Model

$$C_{t+1} = C_t - I \cdot \Delta t \tag{4}$$

Temperature Model

$$T_{t+1} = T_t + k_1 I - k_2(T_t - T_{ambient}) \tag{5}$$

These models were selected to balance physical accuracy with computational simplicity, enabling real-time simulation.

C. AI-Enabled Decision Engine: The core component of the system is an AI-enabled rule-based expert system that performs real-time battery selection. The decision process evaluates multiple state variables:

- Load current
- State of charge (SOC)
- Battery voltage
- Temperature
- Mission phase

The decision logic is structured as a priority-based hierarchy as illustrated in **Fig. 2**.

This structured approach ensures that safety-critical conditions are prioritized over performance optimization by systematically evaluating multiple system parameters under dynamic operating conditions. As shown in **Fig. 2**, the hierarchical flow of decisions enables systematic and interpretable battery selection under varying operational conditions.

D. Intelligent Decision Features: The decision engine incorporates several AI characteristics:

- **Context-aware decision-making:** Simultaneous evaluation of multiple system parameters
- **Mission-aware control:** Adaptive behavior based on flight phase

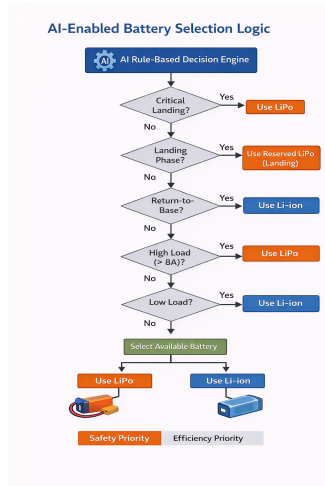

Fig. 2. AI-Enabled Battery Selection Logic

- **Predictive decision-making:** Early initiation of return-to-base based on SOC thresholds
- **Reserve management:** Preservation of LiPo energy for landing operations
- **Hysteresis-based switching:** Prevention of rapid oscillations between battery states
- **Multi-constraint optimization:** Balancing performance, safety, and energy efficiency

This framework functions as an interpretable expert system suitable for real-time embedded applications.

E. Safety and Constraint Handling: To ensure safe operation, the system enforces multiple constraints:

- Voltage cut-off limits for both battery types
- Maximum temperature thresholds to prevent overheating
- SOC thresholds for return-to-base and emergency landing

Emergency landing is triggered when combined SOC falls below a critical limit or when both batteries are depleted. These mechanisms ensure robustness under adverse conditions.

F. Real-Time Visualization and Monitoring: A graphical dashboard is implemented to provide real-time monitoring of system variables, including:

- Battery voltage, capacity, and SOC
- Active battery selection
- Load current and flight time
- Temperature and warning indicators
- Drone position and motion

This visualization enables validation of system behavior and enhances interpretability of the AI decision process.

G. Implementation and Reproducibility: The system is implemented in MATLAB using a modular architecture. Each component—flight simulation, battery modeling, decision logic, and visualization—is independently defined and integrated through a centralized simulation loop.

All parameters, including battery specifications, thresholds, and environmental conditions, are configurable. This ensures that the system can be easily reproduced, extended, and adapted for further research or real-world applications.

Fig. 3. Real-time telemetry dashboard showing battery states, load current, voltage profiles, state of charge, and temperature during simulation.

4. EXPERIMENTS AND RESULTS

This section presents the experimental setup and results obtained from the simulation of the proposed AI-enabled hybrid power management system.

4.1. Experimental Setup

The system is implemented in MATLAB using a discrete-time simulation over a duration of 3600 seconds. A realistic flight profile is generated, including phases such as takeoff, climb, cruise, and maneuvering. The load current varies dynamically between approximately 2 A and 20 A based on flight conditions.

Environmental effects are incorporated using a stochastic wind model, introducing variability in power demand. The battery configurations used are a 2000 mAh 3S LiPo battery and a 1500 mAh 3S Li-ion battery.

The system monitors key variables including state of charge (SOC), voltage, temperature, load current, and active battery selection. Performance is evaluated using SOC evolution, voltage profiles, temperature variation, and battery switching behavior.

4.2. Results

The simulation results are presented using time-series plots of key system variables. The SOC profiles illustrate the distribution of energy usage between LiPo and Li-ion batteries across different flight phases. Voltage curves reflect the nonlinear discharge characteristics of both battery types.

Temperature variation is observed in response to changing load conditions. The battery selection profile shows dynamic switching behavior based on system states and mission phases.

Mission-level outcomes, including return-to-base initiation and emergency landing conditions, are also recorded. The overall system behavior and real-time telemetry outputs are illustrated in **Fig. 3**, providing a comprehensive view of battery states, load dynamics, and system performance during simulation.

All results are visualized through the telemetry dashboard and corresponding plots, enabling comprehensive analysis and observation of system behavior.

5. DISCUSSION

The results show that the proposed AI-enabled hybrid system effectively leverages the strengths of both LiPo and Li-ion batteries. Li-ion is predominantly used during low-load phases, improving energy efficiency, while LiPo is utilized during high-load and landing phases, ensuring reliable power delivery. This behavior aligns with the objective of adaptive and safe energy management.

The priority-based decision framework ensures that safety-critical conditions, such as landing and low energy scenarios, are handled before performance optimization. The reserve strategy for LiPo and the return-to-base mechanism contribute to improved operational reliability.

However, trade-offs are observed. Li-ion batteries, while efficient, cannot support high current demands, whereas LiPo batteries discharge faster and exhibit higher temperature rise under load. The system balances these trade-offs through adaptive switching.

The rule-based decision approach is interpretable and efficient but lacks learning capability. Additionally, the simulation uses simplified battery models and does not account for real-world uncertainties such as hardware constraints and sensor noise.

Overall, the system meets the key objectives of efficient, safe, and adaptive power management, while highlighting opportunities for further improvement.

6. CONCLUSION AND FUTURE WORK

Conclusion

The key contributions of this work are:

- Development of an AI-enabled hybrid power management system integrating LiPo and Li-ion batteries for adaptive energy utilization in FPV quadcopters
- Implementation of a rule-based expert system for real-time, context-aware decision-making using multiple system parameters
- Modeling of nonlinear battery behavior using mathematical and physics-inspired formulations for SOC, voltage, and temperature
- Design of mission-aware and safety-critical logic, including reserve energy management, return-to-base, and emergency landing mechanisms
- Integration of a real-time telemetry and visualization framework for monitoring, interpretability, and system validation
- Establishment of a unified framework combining AI principles, system modeling, and control logic for autonomous UAV energy management

Overall, the system demonstrates that interpretable AI-driven approaches can effectively address real-time decision-making in safety-critical and resource-constrained UAV environments.

Future Work

Future extensions of this work may include:

- Integration of advanced AI/ML techniques (e.g., reinforcement learning, predictive modeling) for adaptive decision-making
- Deployment and validation on real UAV hardware platforms
- Incorporation of advanced battery models including aging, degradation, and efficiency variations

- Extension to multi-battery and multi-drone systems for scalable energy management
- Enhancement of environmental modeling to include sensor noise and uncertainties
- Development of learning-based optimization frameworks using operational data

7. ARTIFACTS AND DEMONSTRATIONS

The following artifacts are provided to support reproducibility and demonstrate the implementation of the proposed system:

- **Code Repository:**
  https://github.com/sagar-sant/AI-Hybrid-Battery-Management-FPV

  Provides the complete MATLAB implementation of the AI-enabled hybrid battery management system, including simulation, decision logic, and visualization modules.
- **Video Demonstration:**
  Watch Simulation Demo

  https://youtu.be/2dLawGw9WpI?si=t07n6XwDi3U6o1FZ

  Demonstrates real-time system operation, battery switching behavior, telemetry dashboard, and mission scenarios.
- **Video Presentation:**
  Watch Video Presentation

  https://youtu.be/NGs5OLFHXAs?si=-aQCaOcIXmHFcXd4

  Includes detailed explanation of system design, methodology, and results.

All artifacts are publicly accessible (or shared with appropriate permissions) and reflect the original work carried out in this project.

8. REFERENCES

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